

Full Alignment Changes for Continuous-Flow Intersection Management

REQ-12 (Replace Existing Text)

The controller shall continuously adjust reservation schedules and vehicle speed profiles to ensure that all vehicles maintain forward movement through the control zone and intersection while preventing collisions and critical-zone conflicts, including during emergency-vehicle priority operations.

REQ-14 (New Requirement)

When an emergency vehicle requests access, the controller shall assign the earliest feasible conflict-free reservation schedule to the emergency vehicle and dynamically reschedule other vehicles without causing collisions or requiring complete stops.

Required Global Changes Throughout the Document

1. Remove all references to 'stop before the intersection'.
2. Replace every 'Command Stop' action with 'Recalculate Reservation and Speed Profile'.
3. Remove STOP from the speed-state model.
4. Keep only FAST, NORMAL, and SLOW operating modes.
5. State explicitly that vehicles may reduce speed but must maintain forward motion.
6. Emergency vehicles receive highest priority reservation.
7. Existing reservations are dynamically rescheduled rather than cancelled.
8. No two vehicles may occupy the same critical zone during overlapping time intervals.
9. Deadlock situations shall be prevented through reservation rescheduling.
10. Every vehicle shall eventually receive a valid conflict-free reservation.

Controller Algorithm Update

Old Logic:

Conflict Detected → Stop Vehicle

New Logic:

Conflict Detected → Recalculate Reservation → Adjust Speed Profile → Continue Moving

Verification Properties for UPPAAL

- Collision freedom must always hold.
- Emergency vehicles must obtain highest priority.
- Vehicles must possess a valid reservation before entering a critical zone.
- Deadlocks are not allowed.
- Vehicles maintain forward motion through adaptive speed control.